

Explainability is Relational: A Receiver-Dependent Framework for AI Interpretability

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https://github.com/RedaAZIZI/impact_social

Abstract

Explainability in AI is treated as an intrinsic property of the model. We argue that this is a category error: explainability is a property of the *relation* between a model M and a receiver R , each endowed with its own representational basis. Classical XAI (feature attributions, surrogate trees) is the degenerate case in which the bases of M and R are assumed identical. We propose (i) a formal definition: M is explainable for R if and only if there exists a decision graph, expressed in R 's predicate vocabulary and of bounded complexity, that simulates M to fidelity ϵ ; (ii) a contrastive, composable *why-operator* W that turns explanation into an interactive protocol whose cost (number of questions) measures explainability; (iii) controlled experiments on a synthetic world whose explanatory ground truth is known. Results: the cost of explainability grows monotonically with the misalignment angle between bases (fidelity AUC $0.918 \rightarrow 0.745$); reconstructing M 's graph through successive why-questions converges in ~ 6 questions under an aligned basis (fidelity 97.9%) and fails to converge under a misaligned one (plateau at 71%) — same model, same answers: only the receiver's basis changes. On real data, the framework predicts where natural misalignment appears: absent on a credit-scoring task whose expert concepts do not carry the signal (a clean, pre-registered negative result), significant on a predictive-maintenance benchmark whose failure laws are written in compound physical quantities ($+0.046$, $p < 10^{-3}$) — and modulated by the internal basis of the model itself. Finally, under continuous drift, a graph maintained through dialogue (one sentence per drift, zero labels) outperforms a network retrained on 75,000 examples, with no cumulative forgetting: we conjecture learning by graph editing as a candidate post-gradient paradigm.

1 Introduction: the hidden assumption of XAI

Between humans, explanation works because a common ground — language, concepts, culture — is shared by default. That channel was never formalized because it never failed. With AI systems the assumption collapses: the internal features of a neural network do not coincide with human concepts. Yet the standard explainability toolbox — SHAP [1], LIME [2], saliency maps, surrogate trees [3, 4] — implicitly assumes that the axes of the model *are* the concepts of the receiver. These methods compute an explanation in the basis of M and deliver it, unchanged, to R .

Thesis. *Explainability is not a property of M . It is a property of the triple (M, R, Q) : the model, the receiver's representational basis, and the contextual question being asked. Classical XAI is the special case $\text{base}(M) = \text{base}(R)$ with $Q = \text{"why this output?"}$ and no explicit contrast.*

This paper is a position paper with preliminary evidence. Our contribution is threefold. First, a formal framework in which explainability is defined relative to a receiver and a question (Section 2). Second, controlled experiments on a synthetic world whose explanatory ground truth is known by construction, showing that the cost of explanation is governed by the alignment between the bases of M and R — not by M alone (Section 3). Third, two consequences of taking the relation seriously: on real data, the framework predicts *where* natural misalignment appears, and we confirm both the positive and the negative prediction (Section 4); and because explanation is a channel, the same channel can be used in reverse — we show that editing an explainable graph through dialogue outperforms gradient-based adaptation under drift, and we state the corresponding conjecture explicitly (Section 5).

If explanation is an interactive protocol — a series of “why?” questions — then the same channel serves learning: the inverse of the why-operator inserts graph fragments supplied by R into M 's explainable graph. The long-term perspectives are (a) AI that learns in dialogue by graph editing — local, instantaneous, traceable — rather than by gradient descent; (b) contextual compression: a large model distilled on the fly, per context, into a small, locally

executable, fully explainable graph; (c) personal AI that grows with its user. The performance–interpretability trade-off [5] is real but is usually evaluated on static benchmarks; in an interactive regime, a graph correctable in one sentence can beat a frozen network that requires retraining. This is a falsifiable thesis, and Sections 3–5 are its first controlled test.

2 Formal framework

2.1 Definitions

- **Receiver.** $R = (V_R, C_R)$: a vocabulary of predicates V_R (the tests R can evaluate and understand) and a capacity constraint C_R (maximal structure size).
- **Candidate explanation.** A decision graph G whose nodes are predicates of V_R , whose edges carry the conditional flow, and whose leaves are outputs.
- **Explainability.** $\text{Expl}(M, R, \epsilon) = \min\{|G| : G \text{ expressed in } V_R, \text{fid}(G, M) \geq 1 - \epsilon\}$, where fidelity is agreement with M on the data distribution. M is explainable for R at level (ϵ, K) iff $\text{Expl}(M, R, \epsilon) \leq K$.
- **Explainability curve.** The maximal fidelity attainable as a function of the budget $|G|$ — the explainability signature of M relative to R . Its area under the curve (AUC) is the associated scalar metric.
- **Degenerate case.** If V_R consists of threshold predicates on M ’s input features (the identity basis), the definition reduces to classical tree distillation: standard XAI is generalized explainability restricted to perfect alignment.

Two axes of complexity interact: structural complexity (the size of the graph) and predicate complexity (the vocabulary V_R). A small graph with dense predicates ($w \cdot x > \theta$) is not readable; an axis-aligned graph is. The “receiver’s basis” is V_R .

2.2 The contextual question Q

Following van Fraassen’s pragmatic theory of explanation [6], every why-question is contrastive (“why A rather than B ?”) and context selects the contrast; see also Miller [7]. In our framework, the answer to Q is the minimal subgraph where the paths toward A and toward B diverge (the pivotal nodes). Counterfactual explanations [8] are a special case. Full explainability then becomes an expectation (or worst case) over the distribution of questions Q of the usage context.

2.3 The why-operator W

$W(x, \text{contrast}) \rightarrow$ minimal pivotal subgraph, expressed in V_R . Three required properties: **minimality** (no superfluous node), **translatability** (output in the receiver’s vocabulary), **composability** (answers assemble). Explanation becomes a protocol: R poses successive W -queries until sufficient simulatability [9, 10] is reached — the “ah, I see” formalized. This yields a second measure of explainability: *the number of W -questions needed to reconstruct M to fidelity ϵ* . The theoretical anchor is Angluin’s exact learning from queries [11] — W is a richer query than membership or equivalence. The connection to interactive proofs and debate [12], with the receiver as a bounded verifier, is left as a one-line perspective.

2.4 Related work

Representational alignment is measured by CKA [13], Procrustes analysis, and model stitching [14]; we use controlled rotations to *manipulate* alignment rather than measure it. Simulatability as the operational test of explanation quality goes back to Doshi-Velez and Kim [9] and Hase and Bansal [10]. Tree distillation of neural networks (TREPAN [3], soft decision trees [4]) is, in our terms, the aligned degenerate case. Rudin [5] argues for interpretable models in high-stakes settings; our framework makes her trade-off receiver-dependent. Catastrophic forgetting [15, 16] and its mitigations such as EWC [17] motivate Section 5: local graph edits do not mitigate the problem, they remove it by construction. Model editing in LLMs (ROME [18], MEMIT [19]) edits opaque weights through fragile localization heuristics; here we edit an explainable structure through the very channel of explanation — editing is a by-product of the framework, not an ad hoc technique. Machine teaching [20] anticipates the data-efficiency of a semantic teaching channel. Finally, a ReLU network is exactly an exponential-size decision graph, so the question becomes one of compressibility — a Kolmogorov-style analogy restricted to graphs.

Table 1: Experiment 1 — fidelity to M as a function of leaf budget and receiver misalignment θ (10 seeds, mean $\pm$ std). Same model, same information (rotations are bijections); only the receiver’s basis varies.

Budget	$\theta=0$	$\theta=0.2$	$\theta=0.4$	$\theta=0.8$	$\theta=1.5$
2	0.696 $\pm$ 0.015	0.678 $\pm$ 0.013	0.654 $\pm$ 0.014	0.595 $\pm$ 0.030	0.549 $\pm$ 0.041
4	0.834 $\pm$ 0.011	0.794 $\pm$ 0.026	0.737 $\pm$ 0.035	0.678 $\pm$ 0.030	0.641 $\pm$ 0.035
8	0.936 $\pm$ 0.009	0.895 $\pm$ 0.014	0.835 $\pm$ 0.014	0.736 $\pm$ 0.017	0.705 $\pm$ 0.023
16	0.976 $\pm$ 0.007	0.934 $\pm$ 0.012	0.879 $\pm$ 0.014	0.791 $\pm$ 0.018	0.752 $\pm$ 0.020
32	0.977 $\pm$ 0.007	0.944 $\pm$ 0.010	0.904 $\pm$ 0.015	0.829 $\pm$ 0.018	0.795 $\pm$ 0.016
64	0.975 $\pm$ 0.008	0.949 $\pm$ 0.007	0.917 $\pm$ 0.013	0.852 $\pm$ 0.014	0.824 $\pm$ 0.019
128	0.975 $\pm$ 0.008	0.952 $\pm$ 0.008	0.920 $\pm$ 0.012	0.872 $\pm$ 0.014	0.841 $\pm$ 0.017
256	0.975 $\pm$ 0.008	0.950 $\pm$ 0.008	0.921 $\pm$ 0.010	0.878 $\pm$ 0.013	0.855 $\pm$ 0.014
AUC	0.918	0.887	0.846	0.779	0.745

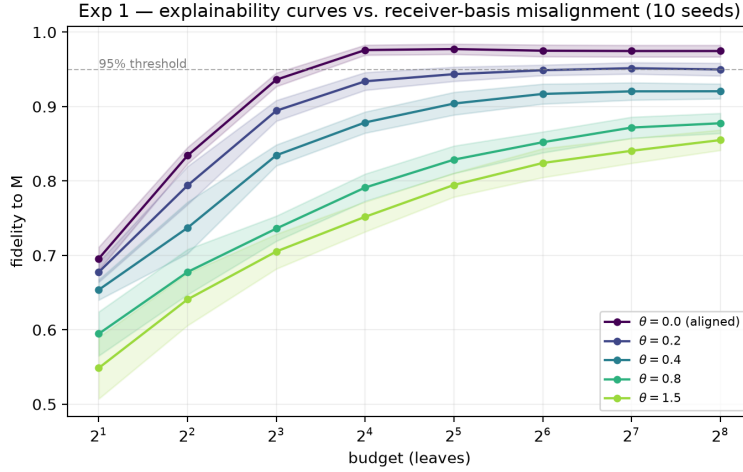


Figure 1: Experiment 1 — explainability curves (fidelity vs. budget, bands $\pm$ std over 10 seeds) for receivers at increasing basis misalignment θ . Not a single weight of M changes across curves.

3 Controlled experiments: the receiver in the equation

All experiments in this section use a synthetic world where the explanatory ground truth is known by construction — the only setting in which the receiver’s basis can be manipulated exactly. The world has 6,000 objects with 5 interpretable attributes $\sim U[0, 1]$ (size, hue, shape, position x , position y) and a hand-written ground-truth rule graph G^* with 5 branches (e.g. *if size > 0.6 and hue > 0.5 $\rightarrow$ class A*). The opaque model M is an MLP (2×50 units) trained on $(X, G^*(X))$, reaching accuracy 0.9762 ± 0.0059 (10 seeds). Code, seeds, and figures are in the repository (Section 7).

3.1 Experiment 1: the cost of explanation grows with basis misalignment

Protocol. Receivers are defined by their basis: a rotation of strength θ applied to the 5 dimensions ($R_\theta = \exp(\theta A)$ with A a normalized random antisymmetric matrix; $\theta = 0$ is the identity), $\theta \in \{0, 0.2, 0.4, 0.8, 1.5\}$. Explainability is measured by distilling M into trees with threshold predicates *in the receiver’s basis*, for leaf budgets 2 to 256; fidelity is agreement with M on held-out data; 10 seeds, mean $\pm$ std. The falsification criterion was set a priori: superposed curves would void the concept.

Results (Table 1, Figure 1). The explainability AUC decreases strictly monotonically with θ : 0.918 at $\theta=0$ down to 0.745 at $\theta=1.5$. The minimal budget for fidelity ≥ 0.95 is 16 leaves at $\theta=0$, 128 at $\theta=0.2$, and is never reached (>256) for $\theta \geq 0.4$.

Interpretation. Same model M (not one weight changes), same information (rotations are bijections). Only the receiver’s basis varies, and the explainability curves diverge monotonically with θ . Explainability is therefore not a property of M alone.

Table 2: Experiment 2 — fidelity to M of the reconstructed model vs. number of why-questions (5 seeds). Same M , same answers to every query; only the receiver’s basis differs.

Questions	R aligned	R rotated ($\theta=1.5$)
0	0.457 ± 0.010	0.457 ± 0.010
3	0.727 ± 0.089	0.625 ± 0.046
6	0.921 ± 0.022	0.688 ± 0.024
9	0.966 ± 0.002	0.695 ± 0.021
12	0.977 ± 0.006	0.703 ± 0.027
15	0.979 ± 0.007	0.713 ± 0.024

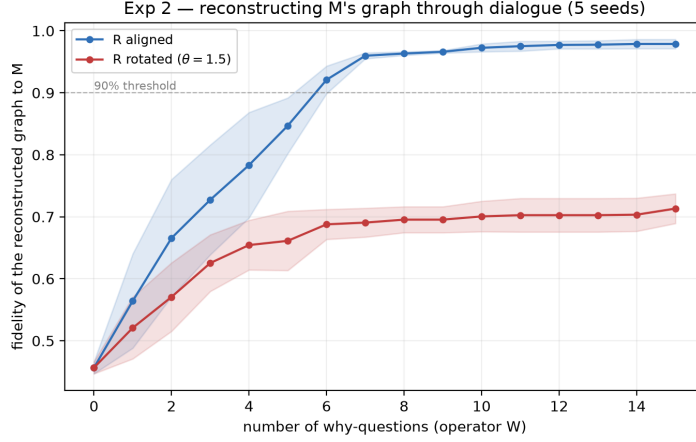


Figure 2: Experiment 2 — reconstruction of M by successive why-questions. The aligned receiver reaches 90% fidelity in 6 questions; the misaligned receiver plateaus near 71% with the same answers from M .

3.2 Experiment 1b: what costs is mixing, not reparametrization

Applying a coordinate-wise monotone deformation on top of the rotation ($v = \tanh(s \cdot (x - 0.5) \cdot R_{\theta=1.5})$, $s \in \{2, 6\}$, 5 seeds) leaves the AUC unchanged: 0.7432 vs. 0.7433 vs. 0.7433 for the linear, $s=2$, and $s=6$ cases. For a receiver with threshold predicates, the cost of explainability is *invariant under coordinate-wise monotone deformations* of its basis — a threshold transports through any monotone function. What costs is the *mixing* of dimensions, not their reparametrization: the relevant misalignment is structural, not metric. This invariance was subsequently confirmed on every test bed of this paper, synthetic and real (11 runs out of 11, often at exactly 0.0000); a formal proof (threshold trees depend only on coordinate orders) is left to future work.

3.3 Experiment 2: explanation as a measurable protocol

Protocol. Same world, same M ; the reference graph of M is a 32-leaf tree distilled in the original basis. The operator $W(x)$ answers “why is x classified c ?” with the decision path of x (a conjunction of threshold predicates) plus its class — minimality via the path, implicit contrast versus the other leaves. Dialogue protocol (active learning): the receiver starts from a trivial majority-class predictor; at each turn it picks a point where it disagrees with M , asks W , integrates the rule, and repeats (15 questions max, 5 seeds). The *aligned* receiver stores the rule verbatim; the *rotated* receiver ($\theta=1.5$) perceives the world only in its own basis, so each received rule — an axis-aligned region for M , an oblique polytope for it — must be approximated by a small 8-leaf tree in *its* vocabulary.

Results (Table 2, Figure 2). The aligned receiver reaches 90% fidelity in 6 questions and 97.9% at 15; the rotated receiver never reaches 90% and plateaus at $\sim 71\%$. Structurally, G^* has ~ 7 branches and ~ 6 questions suffice: the dialogue reconstructs the structure at roughly one rule per question.

Interpretation. Explainability-as-protocol is measurable: the number of why-questions is small and finite when bases are aligned, and the dialogue fails to converge when they are misaligned — *with identical answers from M* . Two independent protocols (static distillation, interactive dialogue) confirm the same thesis.

Table 3: Experiment 5 (German Credit) — expert vs. raw vocabulary, $n = 10$ seeds, paired per-seed statistics. The pre-registered falsification criterion fired: a clean negative result.

Run	AUC raw	AUC expert	Paired gap	Paired t	Wilcoxon
v0 (MLP)	0.733 $\pm$ 0.025	0.727 $\pm$ 0.019	−0.006 $\pm$ 0.018	$p = 0.32$	$p = 0.48$
v0.1 (GBT + queries)	0.803 $\pm$ 0.026	0.801 $\pm$ 0.021	−0.002 $\pm$ 0.019	$p = 0.72$	$p = 0.83$

3.4 Experiment 4: explainability depends on the question

On the reference graph of M , we measure the size of the contrastive answer to “why a rather than b ?” — the minimal number of diverging predicates between the instance’s path and the closest path leading to b , weighted by leaf coverage. The answer size ranges from 2.95 predicates (B rather than A) to 6.39 (B rather than C), and the relation is asymmetric: A-vs-B costs 5.81 while B-vs-A costs 2.95. The same model is twice as “explainable” for some questions as for others, consistently with the pragmatic theory: explanation is not symmetric in the contrast. Full explainability should thus be defined as an expectation over the question distribution of the usage context, $\text{Expl}(M, R, \bar{Q}) = \mathbb{E}_Q[\text{answer size}]$.

4 Contact with real data: a predictive diptych

If misalignment is real, it should occur *naturally*: some domains describe the world in compound expert concepts, others do not. The framework makes a falsifiable prediction about where an expert vocabulary should dominate a raw one — and we tested it on two public datasets with pre-registered hypotheses and falsification criteria, reporting the negative result with the same prominence as the positive one. Both experiments use $n = 10$ seeds and per-seed paired statistics (paired t -test and Wilcoxon signed-rank), and both include the monotone control of Section 3.2 (log/power transforms of raw features), predicted inert.

4.1 Experiment 5 (negative): German Credit

Protocol. Statlog German Credit [21]: 1,000 applications, 20 attributes, binary good/bad outcome (70/30). Receiver vocabularies: *raw* (features as-is); *expert* = raw + compound credit concepts (installment = amount/duration, burden = installment $\times$ repayment rate, exposure = amount $\times$ number of credits); monotone control. Registered protocol v0 used an MLP (2×50); a documented amendment v0.1 switched to gradient boosting plus 20,000 extra distillation queries after the v0 MLP (accuracy 0.72 against a 0.70 majority class) turned out to have almost no structure to explain — the methodological lesson, incorporated into all subsequent protocols, is to normalize the explainability curve by the fidelity of the trivial model. Falsification criterion set a priori: an expert–raw AUC gap below one inter-seed standard deviation means no natural misalignment on these data.

Results (Table 3). No dominance of the expert vocabulary, in either run. The monotone control is inert as predicted (−0.0008 $\pm$ 0.0016 in v0; 0.0000 in v0.1) — the invariance of Section 3.2 holds on real data.

Interpretation. The falsification criterion spoke: no natural misalignment on German Credit. Two readings: (a) the predictive signal of this dataset lives in categorical variables (account status, credit history) shared by both vocabularies, and the central expert concept of credit — the debt-to-income ratio — is not computable from the available attributes (no income); (b) more deeply, expert-vocabulary dominance requires the world’s generating process to be itself written in compound quantities. That is precisely what Experiment 6 tests.

4.2 Experiment 6 (positive): predictive maintenance

Protocol. AI4I 2020 Predictive Maintenance [22]: 10,000 points, 5 sensors (air and process temperatures, rotational speed, torque, tool wear). The benchmark’s failure modes are *written* in compound quantities: heat dissipation (ΔT and speed), power (torque $\times$ angular speed), overstrain (wear $\times$ torque). Hypothesis registered a priori: on physical ground — where the generating process is written in compound quantities — the engineer’s vocabulary (sensors + ΔT , power, strain) dominates the raw-sensor vocabulary; same falsification criterion as Experiment 5. Fidelity is class-balanced (failures are rare — the lesson of Experiment 5), budgets 2 to 256 leaves, $n = 10$ seeds, paired tests, and two opaque models: an MLP (2×50) and gradient boosting.

Results (Table 4, Figure 3). For the MLP, the engineer’s vocabulary dominates (+0.046 $\pm$ 0.026, paired $t = 5.37$, $p = 4.5 \times 10^{-4}$, Wilcoxon $p = 0.002$). For gradient boosting, the gap vanishes (−0.005 $\pm$ 0.027, $p = 0.56$). Monotone

Table 4: Experiment 6 (AI4I 2020) — engineer vs. raw vocabulary, $n = 10$ seeds, paired per-seed statistics, class-balanced fidelity. The gap depends on the internal basis of M itself.

M	AUC raw	AUC engineer	Paired gap	Paired t	Wilcoxon
MLP	0.702 ± 0.050	0.748 ± 0.048	$+0.046 \pm 0.026$	$t = 5.37, p = 4.5 \times 10^{-4}$	$p = 0.002$
GBT	0.760 ± 0.022	0.755 ± 0.017	-0.005 ± 0.027	$p = 0.56$	$p = 0.56$

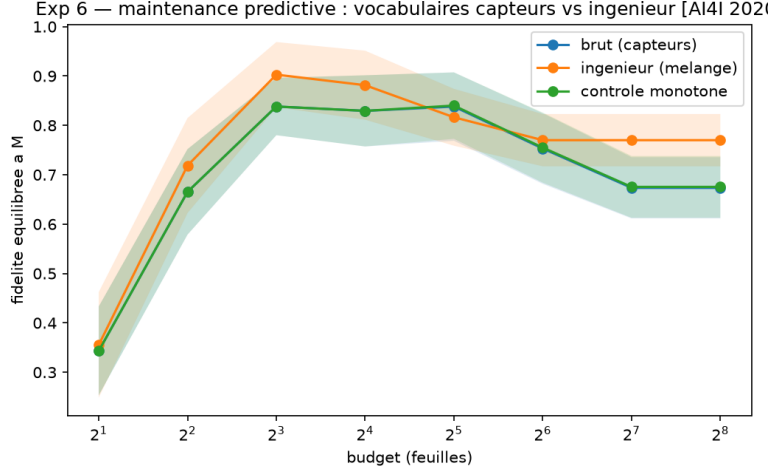


Figure 3: Experiment 6 — explainability curves on AI4I 2020 for raw vs. engineer vocabularies (class-balanced fidelity, 10 seeds). The engineer’s vocabulary dominates for the MLP; the gap vanishes for the axis-aligned GBT.

controls are inert in both cases. Balanced accuracy of M is 0.62 (MLP) and 0.71 (GBT); the residual failure modes of AI4I are known to be nearly unlearnable.

Interpretation. To our knowledge, this is the first controlled demonstration of *natural* basis misalignment on real data. Together with Experiment 5 it forms a diptych: no dominance where the world is not written in compound concepts (credit), clear dominance where it is (physics) — natural misalignment exists, and the framework predicts where. The GBT result is the structural surprise: a tree ensemble is built from axis-aligned splits on raw features — *its internal basis is the raw vocabulary by construction* — and the gap disappears; the MLP mixes dimensions the way the physics does, and the gap appears. Same data, same receivers, different M , different explainability relation: the relational thesis observed *on the model side*, everything else held fixed. Caveats: one dataset per verdict; expert vocabularies chosen by the authors, not learned; rare classes limit the learnable structure of M .

5 The interactive regime: the channel that explains is the channel that learns

The why-operator has an inverse: where W extracts a rule from M ’s explainable graph, W^{-1} inserts or edits one, supplied by the receiver in the shared vocabulary. Because an edit touches exactly the rules it names, regions outside the edit are intact *by construction* — catastrophic forgetting [15, 16] is not mitigated, it is removed. This section tests the resulting learning regime, first on a single correction, then under continuous drift.

5.1 Experiment 3: one sentence vs. thousands of examples

Protocol (concept drift). The world changes: the rule “shape $> 0.7 \rightarrow B$ ” of G^* becomes “shape $> 0.7 \rightarrow A$ ”. M , trained on the old world, drops to 0.741 accuracy. Route A (our framework): the explainable graph of M (rule list extracted from the 32-leaf reference tree) is corrected by *one* sentence of dialogue — “when shape > 0.7 , it is A now” — implemented as W^{-1} editing the class of the concerned rules; cost: one interaction, zero gradients. Route B (standard paradigm): the MLP is fine-tuned (warm start, 50 epochs) on $n \in \{10, \dots, 3000\}$ freshly labeled examples. Accuracy on the new world, 5 seeds.

Table 5: Experiment 3 — adaptation to a rule flip: one graph edit through dialogue vs. fine-tuning on labeled examples (5 seeds).

Method	Cost	Accuracy (new world)
M without adaptation	—	0.741 ± 0.012
Graph edit (dialogue)	1 sentence	0.990 ± 0.004
Fine-tuning	10 examples	0.758 ± 0.046
Fine-tuning	100 examples	0.893 ± 0.012
Fine-tuning	1000 examples	0.947 ± 0.004
Fine-tuning	3000 examples	0.968 ± 0.004

Table 6: Experiment 7 — mean accuracy over 15 drifts (5 seeds). The dialogue-maintained graph uses 15 sentences and zero labels; all four pre-registered kill conditions survived.

Method	Total supervision (15 drifts)	Mean accuracy (drifts 1–15)
Frozen MLP	—	0.624 ± 0.002
Fine-tuned MLP	1,500 labels	0.854 ± 0.018 (forgetting sawtooth, min 0.740)
Retrained-from-scratch MLP	75,000 labels	0.981 ± 0.002
Graph + dialogue	15 sentences, 0 labels	0.990 ± 0.003
Graph, noisy sentences ± 0.05	15 sentences	0.984 ± 0.008
Graph, noisy sentences ± 0.10	15 sentences	0.971 ± 0.018
Graph, misaligned basis $\theta = 1.5$	15 sentences	0.801 ± 0.028

Results (Table 5). One sentence of dialogue outperforms 3,000 examples of fine-tuning (0.990 vs. 0.968). The edit is instantaneous, local, and traceable.

Methodological fair play. The comparison is deliberately *not* information-equal: one semantic sentence compresses what thousands of examples express — that is the thesis, not a confound. The explanation channel (W) and the learning channel (W^{-1}) are the same channel used in both directions. Note also that Route A requires the correction to be expressible in the shared vocabulary: under a misaligned basis (Experiments 1–2) this channel degrades, which ties the experiments into a single narrative — and Experiment 7 quantifies exactly that tax.

5.2 Experiment 7: a graph maintained by dialogue under continuous drift

Protocol (pre-registered; kill conditions published before the run — the git record is the evidence). The reference world G^* undergoes 15 successive drifts: at each step the class of one branch flips (deterministic cyclic schedule). Competitors: (a) the explainable graph (32 leaves distilled from the initial M) maintained by dialogue — *one* sentence per drift (“in the region (branch condition), the class is now c ”), interpreted against the receiver’s own unlabeled experience (any rule whose firing region lies mostly inside the sentence’s region is re-classified); zero labeled examples; (b) the frozen MLP; (c) the MLP fine-tuned on 100 fresh labeled examples per drift (50 epochs); (d) the MLP retrained from scratch on 5,000 examples per drift. 5 seeds. Stress variants: noisy sentences (thresholds perturbed by ± 0.05 and ± 0.10 — human imprecision); receiver in a misaligned basis ($\theta = 1.5$). Four kill conditions were set a priori: K1, the graph falls below fine-tuning over the last 5 drifts; K2, accuracy of a never-edited region degrades after 15 edits; K3, the advantage over the frozen model vanishes under ± 0.10 noise; K4, the misaligned graph falls below the frozen model.

Results (Table 6, Figure 4). Fifteen sentences with zero labels outperform 75,000 labels of from-scratch retraining (0.990 ± 0.003 vs. 0.981 ± 0.002). All four kill conditions survived. Integrity of the never-edited region after 15 edits: $0.992 \rightarrow 0.994$ — zero cumulative degradation (K2: locality holds over 15 edits). Human imprecision of ± 0.10 costs only ~ 2 points (0.971). The misaligned receiver pays a measured *expressibility tax* of ~ 0.19 (0.801), quantitatively linking Experiments 1–2 to Experiment 3. An unplanned control: the cyclic schedule returns the world to its initial state at drift 12, where the frozen model climbs back exactly to its original level — the drift mechanics are sound.

The conjecture, stated as such. We conjecture that learning by graph editing in dialogue is a *candidate paradigm for post-gradient learning*: data-efficient because the channel is semantic, and continual by construction because edits are local. It attacks head-on the two best-documented weaknesses of deep learning — data inefficiency (one sentence compresses thousands of examples, Experiment 3) and catastrophic forgetting (locality removes the problem rather than mitigating it, Experiment 7 / K2). This is a conjecture supported by controlled synthetic evidence, not an established claim.

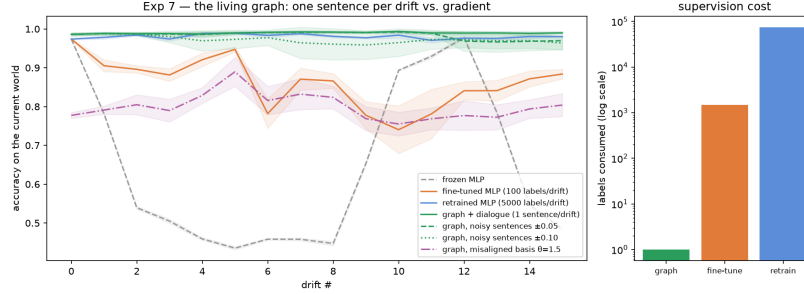


Figure 4: Experiment 7 — accuracy along 15 successive drifts for the four competitors and the stress variants. The dialogue-maintained graph (15 sentences, zero labels) dominates throughout; the never-edited region shows zero cumulative degradation.

6 Limitations

- Synthetic experiments are low-dimensional with G^* known by construction. This is a choice — total control of misalignment, impossible with human subjects — but generalization to high dimension remains to be demonstrated.
- Receivers are artificial. Human validation (experts vs. novices as different bases) is the next step, not the present contribution; no claim in this paper concerns human receivers.
- Misalignment is modeled by linear rotations (plus monotone deformations); real conceptual misalignments are nonlinear and structured.
- The “graph of M ” is a high-fidelity distilled proxy, not the exact network.
- Experiment 3’s dialogue/fine-tuning comparison is not information-equal (assumed — it is the thesis), and identifying the rule to edit is assisted by knowledge of G^* ; in real conditions, localizing the faulty rule is a problem in itself.
- The performance–interpretability trade-off [5]: our synthetic worlds are compressible by construction. The thesis “an interactive graph beats a frozen network under continual learning” is now tested in a synthetic regime (Experiment 7), not yet on real streams.
- Experiments 5–6: one dataset per verdict of the diptych; expert vocabularies are chosen by the authors, not learned; on AI4I, near-random residual failure modes limit the learnable structure of M .
- Experiment 7: sentences come from an oracle that localizes the concerned region (noisy variants tested; autonomous localization unresolved); the receiver has the world’s inductive bias — the $\theta = 1.5$ variant prices the opposite case.

7 Perspectives and conclusion

One line each, in order of priority: human validation of the framework (simulatability measured on subjects, bases operationalized by expertise); autonomous localization of the rule to edit when the correction does not name it; more datasets per verdict of the real-data diptych, with *learned* rather than chosen expert vocabularies; a formal proof of the monotone-invariance proposition and a characterization of the cost by mixing structure (angles between subspaces); contextual compression of large models — “the best graph of size K for context C ” — as a primitive; weighting explainability by realistic question distributions Q ; construction of a common basis through a bilateral feedback loop, superposition of crisp graphs as a model of vague language, and a calibration operator; and the connection to interactive proofs and debate [12], with the receiver as a bounded verifier.

Explainability research keeps asking what a model must look like to be understandable. We propose to ask, instead, *for whom* it is understandable — and to make the receiver a first-class citizen of the definition. Once the receiver is in the equation, explanation stops being a report and becomes a channel: measurable in questions (Experiments 1–2, 4), predictive on real data in both directions (Experiments 5–6, including a clean pre-registered negative result), sensitive to the model’s own internal basis (Experiment 6), and reversible into a learning mechanism whose locality abolishes forgetting by construction (Experiments 3, 7). The evidence presented here is deliberately preliminary — controlled worlds, artificial receivers, honest kill conditions — but the direction is stated without hedging: there is no explainability

without a receiver.

Reproducibility

All experiments (10 or 5 seeds, fixed), the rex library (rule-list model, W , W^{-1} , metrics, with unit tests including the edit-locality invariant), figures, and dataset download scripts are available at https://github.com/RedaAZIZI/impact_social. Experiment 7’s hypotheses and kill conditions were pre-registered in the repository before the run; Experiment 5’s protocol amendment (v0 $\rightarrow$ v0.1) is documented in the same place. No dataset is redistributed; both real datasets are public UCI benchmarks [21, 22].

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